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Q1. A vector space is a set closed under addition and scalar multiplication.

Example: $\mathbb{R}^3$ with the usual operations.

Q2. For $u \cdot v = |u||v|\cos(\theta)$, swapping u and v does not change the value.

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Q3. $(1,2,3) \times (4,5,6) = (2 \cdot 6 - 3 \cdot 5, 3 \cdot 4 - 1 \cdot 6, 1 \cdot 5 - 2 \cdot 4) = (-3, 6, -3)$.

Q4. Vectors are independent if only the zero combination gives the zero vector.

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Q5. $\cos(\theta) = (u \cdot v) / (|u||v|)$, then $\theta = \arccos$ of that value.

Q6. (left blank by the candidate)